Shahjalal University of Science and Technology
Department of Computer Science and Engineering
3rd Year 2nd Semester Final Examination' Dec 2018 (Session: 2015-16)
Course Code: CSE 329 Credits: 3 Course Title: Computer Architecture
Time: 3 hrs Total Marks: 100

Group A

[Answer all the questions]

1. Determine if the following statements are true or false. Answer any TEN. **10x1=10**

a) Spatial locality states two instructions with nearby addresses are referenced close together in time.
b) MIMD targets Task-level parallelism.
c) Clock rate is stabilised to limit power consumption.
d) Slowing clock rate reduces energy.
e) Overclocking a processor reduces power consumption.
f) Service interruption is measured with MTTF.
g) SPECWeb is used for desktop benchmark.
h) Write stall is not affected by the write-back strategy.
i) Virtual address complicates program relocation.
j) Cache operations are dominated by read operations.
k) If a block can be placed anywhere in the cache, the cache is said to be fully associative.
l) MISD means many instruction single data.
m) CPU clock cycle for a program is inversely related to IPC.
n) Pipelining reduces the instruction time of an individual instruction.
o) Data hazard arises because of dependencies on consecutive instructions.

2. Answer any FOUR **4x5=20**

a) i. Write a few characteristics of an embedded computer. 2
ii. Differentiate between Thread-level and Request-level parallelisms. 3

b) i. Explain MIPS register addressing with examples. 3
ii. What are the key properties of a flash memory? 2

c) i. Suppose a program takes 1 billion instructions to execute on a processor running at 4 GHz. Suppose also that 20% of the instructions execute in 3 clock cycles, 30% execute in 4 clock cycles, 40% execute in 4 clock cycles and 10% execute in 5 clock cycles. What is the execution time for the program or task? 3
ii. What is the critical word first mechanism? 2

d) Explain the effect of availability, scalability and throughput between server and desktop computers. 5

e) i. Explain the phenomenon hit-under-3-misses. 3
ii. How do hit time, miss rate and miss penalty affect average memory access time? 2

f) i. Assume a fully associative write-back cache with many cache entries that starts with an address of 400. Below is a sequence of five memory operations (the address is in square brackets) 3

- Read Mem[400];
- Write Mem[500];
- Read Mem[100];
- Write Mem[400];
- Read Mem[500].

How many hit and miss will occur for write allocate and no-write allocate?
ii. Compare and contrast between SRAM and DRAM. 2

3. Answer any TWO **2x10=20**

a) i. Assume that a 25% reduction in frequency results in a 30% reduction in voltage. What would be the impact on dynamic energy and dynamic power? 4
ii. What is a memory stall cycle? How does it impact the CPU execution time of a program? 3
iii. Explain the purposes of PRE and ACT commands in DRAM. 3

b) i. Show that the ratio of the geometric means is equal to the geometric mean of the performance ratios. 4
ii. What are the two states of a service with respect to an SLA? How and when there is a transition between these two states? 4
iii. What causes the RAW (Read-After-Write) hazard? 2

c) i. What is the purpose of utilizing machine memory? 3
ii. Briefly explain the purpose of instruction fetch, memory access and write-back cycles. 5
iii. Explain the usefulness of static branch prediction. 2

Group B

[Answer all the questions]

4. **Answer any TEN** **10x1=10**

a) What is an ILP?
b) What is the basic property of load-store ISA?
c) What is feature size?
d) Why energy is a better metric than power?
e) What happens in a no-write allocate?
f) How miss per instruction and miss rate are related?
g) What is the write through strategy?
h) What is the cause of static power?
i) What do you mean by yield?
j) What does FIT measure?
k) State Amdahl's law.
l) What is an out-of-order execution?
m) What is block identification?
n) In terms of hit ratio and search speed which one is better for direct mapped and fully associative cache?
o) What are control hazards?

5. **Answer any FOUR** **4x5=20**

a) i. Differentiate between SIMD and MIMD. **2**
ii. What is the effect of write buffer? **3**

b) i. Determine the MTTF for 10^7 FIT. **2**
ii. Explain the effect of slowing clock rate and lowering voltage over dynamic power and dynamic energy **3**

c) Compare and contrast the effect of larger block size and bigger cache to reduce miss rate. **5**

d) Briefly explain the performance improvement over the years among memory, processor and disk storage in terms of bandwidth and latency. **5**

e) Explain the phenomenon cache hit, cache miss, page fault and page table. **5**

f) Explain how way prediction can reduce hit time. **5**

6. **Answer any TWO** **2x10=20**

a) i. Compare and contrast among compulsory, capacity and conflict misses. **6**
ii. Explain how data is located in the cache. **4**

b) i. How merging write buffer can be leveraged to reduce miss penalty? **4**
ii. Explain how the principal of locality and spatiality can increase the performance of a computer. **4**
iii. Explain why MTTF is a reliability measure. **2**

c) i. Briefly explain the classic five-stage pipelining for a RISC processor. **6**
ii. Differentiate between structural and control hazards. **4**